Multilevel Regression for Spawner Surveys

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Introduction

> ***Let the data speak for itself... but all the data must speak.***
> - John P. Cunningham (2012)

Estimates of the number of sockeye salmon (*Oncorhynchus nerka* Walbaum 1792) spawners in a river or a lake beach involve repeated surveys during the spawning period. Typically spawning is about one month after the autumnal equinox but timing varies greatly within and between species of *Oncorhynchus*. Techniques vary, conventionally: mark-recapture, and visual counts while snorkeling, walking, or flying in a helicopter or small airplane; recently: analyzing video from drones. There are many other methods to estimate abundance: mark-recapture, conductivity sensors, counting fences, and eDNA. Other salmon species may be counted in a survey for sockeye, and each salmon species has a diversity of enumeration methods.

The analysis of repeated surveys of sockeye spawners is conventionally an area under the curve estimate (AUC) where the "curve" is typically the area under straight lines joining observations versus days (a trapezoid). This analysis is the same as determining the cumulative abundance, again as straight lines between (accumulating) observations. The data may be supplemented with prior knowledge or subjective judgments about first, peak, and last presence of spawners. One improvement assumes a Gaussian ($\mathcal{N}(\mu, \sigma)$, normal) distribution for the temporal pattern of abundance, fitted to a time series survey estimates. These analyses are confined to the samples for one river in one year, so parameter estimates are based on few data, < 10, and abundance estimates are accordingly imprecise.

The intention of this exercise is to show, using simulated data, that AUC estimates can be improved by considering multiple rivers and multiple years in a multi-level regression. The ideas are:

- there will be an effect in common due to years on the temporal pattern of abundance in a group of river. Marine migration might be early or late, rainfall to raise water levels sufficiently for salmon to enter the rivers may be early or late.
- a river can have a consistently different temporal pattern from other river. Spawners that home to that river can arrive earlier or later, and more or less spread in time.
- a river can have consistently more or fewer salmon compared to other rivers. Factors such as length of accessible river and the abundance of gravel suitable for spawning (for redds) affect salmon abundance.; and
- the abundance of spawners across all rivers varies by year. Implicit is that these spawning groups in rivers (salmon tribes?) are related within a population or conservation unit (metapopulation).

These ideas can be assembled into an equation

$$N_{tjk} = N_j P_k (T_0[\mu, \sigma]_t + T_j[\mu, \sigma]_t + T_k[\mu, \sigma]_t) + \epsilon_{tjk} \tag{1}$$

where N_{tjk} is the number of spawners on day i in river k in year j. That is estimated with error ϵ from the proportion P_k of all spawners that year N_j which enter river k according to the background temporal pattern T_0 and how that pattern is affected by the pattern for that year T_j and that river T_k. The expression $T_x[\mu, \sigma]_t$ refers to a temporal distribution with parameters μ_x and σ_x for timing and spread (if the normal distribution is chosen).

The normal distribution might be a sufficient description for the temporal abundance of spawners despite being symmetrical and unbounded. Reality might be better described as a truncated and skewed Gaussian distribution, or by a distribution that is inherently bounded. The beta distribution, for example has the advantage that, after determining bounds such as the first and last important abundance of spawners in all rivers (in the region being analyzed) then river effect and year effect might be sufficiently described by a single parameter, the day of peak abundance (mode), that determines the shape of the beta distribution between those bounds (the skew).

```
#|label: setup
#|echo: false
library(rstan); library(magrittr)
```

```
Loading required package: StanHeaders
```

```
rstan version 2.32.6 (Stan version 2.32.2)
```

```
For execution on a local, multicore CPU with excess RAM we recommend calling
options(mc.cores = parallel::detectCores()).
To avoid recompilation of unchanged Stan programs, we recommend calling
rstan_options(auto_write = TRUE)
For within-chain threading using `reduce_sum()` or `map_rect()` Stan functions,
change `threads_per_chain` option:
rstan_options(threads_per_chain = 1)


Attaching package: 'magrittr'

The following object is masked from 'package:rstan':

    extract
```

Examples of Samples

Temporal patterns from year and river effects.

Simulate data applies the normal distribution for the overall abundance of spawners with timing (mean, μ) centered on day 273, the end of September, and a spread (sd, σ) of 7 days. The year effect on the timing (mean) for spawner abundance in one river is drawn from a random normal distribution with sd = 5 days and mean = 0. The year effect on the spread (sd) of the spawner abundance in a river is similar, random normal with sd = 1.5 days and mean = 0. The river effects are the same, random normal with timing sd = 5 days and spread sd = 1.5 days.

The following code creates a three dimensional array of *n_day* observations in each of *n_river* rivers in each of *n_year* years.

```r
#|label: temporal_patterns
# mean and SD of salmon abundance after removing year
# and river effects. Day 270 is September 30 (not leap year)
background <- c(273, 7)
# storage for simulated data
n_day <- n_river <- n_year <- 5;  # remember: arrays stored by first index
trueAbundance <- array(dim=c(n_day, n_river, n_year));
dimnames(trueAbundance) <-list(
    day=1:n_day,river=LETTERS[1:n_river],
    year=1991:(1991+n_day-1) )

# river size effect:proportion across all years, of spawners in that river
```

```r
river_size <- runif(n_river,1,10);   # random uniform.
river_size <- river_size/sum(river_size); # proportion
# temporal effects effects
yearSD <- c(5,1.5) # effect on timing and spread
riverSD <- c(5,1.5)
yearEffect <-  cbind(rnorm(n_year,   0, yearSD[1]),
                     rnorm(n_year,   0, yearSD[2]) ) # 2 columns
riverEffect <- cbind(rnorm(n_river, 0, riverSD[1]),
                      rnorm(n_river, 0, riverSD[2]), river_size) # 3 columns
dimnames(yearEffect) <- list(year=1991:(1990+n_day),
                             effect=c("timing", "spread"));
dimnames(riverEffect) <- list(river=LETTERS[1:n_river],
                             effect=c("timing", "spread", "size"));
cat(  ' Year Effect   \n'); print(yearEffect)
```

```
 Year Effect


      effect
year      timing      spread
  1991 -7.828674 -2.05094866
  1992  2.066193  0.04930031
  1993  1.731602 -1.77898826
  1994 -1.219672  3.02330265
  1995  4.005117 -0.52432543
```

```r
cat('\n River Effect \n'); print(riverEffect)
```

```
 River Effect


     effect
river     timing        spread      size
    A   1.652758  0.4867256616 0.2944752
    B   1.850563  0.1907350341 0.2652708
    C -11.478179 -1.1335223286 0.0585020
    D  10.270845  0.0002778112 0.1114069
    E   3.224785  1.3601720411 0.2703450
```

Examine total effect.

The random, additive effects of rivers (A, B, C,) and year (1991, 1992,) are calculated for timing and spread.

```
b1  <- matrix(nrow=n_river, ncol=n_year);
dimnames(b1) <- list(river=LETTERS[1:n_river], year=1991:(1990+n_day) )
b4 <- b3 <- b2 <- b1
cat("riverEffect+yearEffect. Timing (mean) \n")
```

```
riverEffect+yearEffect. Timing (mean)
```

```
for(year in 1:n_year){
  for(river in 1:n_river){
    b1[river, year]=riverEffect[river,1]+yearEffect[year,1];
  }
}
print(b1)
```

```
     year
river       1991      1992      1993        1994      1995
    A  -6.175917  3.718951  3.384360   0.4330857  5.657875
    B  -5.978111  3.916756  3.582166   0.6308913  5.855681
    C -19.306853 -9.411986 -9.746577 -12.6978510 -7.473062
    D   2.442171 12.337038 12.002448   9.0511732 14.275963
    E  -4.603889  5.290978  4.956388   2.0051131  7.229902
```

```
cat("\n riverEffect+yearEffect. Spread (SD) \n")
```

```
 riverEffect+yearEffect. Spread (SD)
```

```
for(river in 1:n_river){
  for(year in 1:n_year){
    b2[river, year]=riverEffect[river,2]+yearEffect[year,2];
  }
}
print(b2)
```

```
     year
river       1991        1992       1993     1994        1995
    A -1.5642230  0.53602597 -1.2922626 3.510028 -0.03759977
    B -1.8602136  0.24003534 -1.5882532 3.214038 -0.33359040
    C -3.1844710 -1.08422202 -2.9125106 1.889780 -1.65784776
    D -2.0506708  0.04957812 -1.7787105 3.023580 -0.52404762
    E -0.6907766  1.40947235 -0.4188162 4.383475  0.83584661
```

```r
# earliest and last abundance across all rivers and years
# needed to determine sampling days
b3 <- b1 - 1.5*(b2+background[2]);
# first case revealed
cat("\n Early due to river and year effects \n");
```

```
 Early due to river and year effects
```

```r
print(b3);
```

```
     year
river       1991       1992       1993       1994       1995
    A -14.329582  -7.585088  -5.177246 -15.331957  -4.785725
    B -13.687791  -6.943297  -4.535454 -14.690165  -4.143934
    C -25.030147 -18.285653 -15.877811 -26.032521 -15.486290
    D  -4.981823   1.762671   4.170513  -5.984197   4.562034
    E -14.067724  -7.323230  -4.915388 -15.070099  -4.523867
```

```r
b4 <- b1 + 1.5*(b2+background[2]);
# last  case revealed
cat("\n Late due to river and year effects \n");
```

```
 Late due to river and year effects
```

```r
print(b4)
```

```
     year
river       1991       1992      1993       1994       1995
    A   1.977749 15.0229897 11.945966 16.1981282 16.1014754
```

```
B   1.731568 14.7768094 11.699786 15.9519478 15.8552950
C -13.583560 -0.5383189 -3.615342  0.6368195  0.5401667
D   9.866165 22.9114055 19.834382 24.0865439 23.9898911
E   4.859946 17.9051867 14.828163 19.0803252 18.9836724
```

Generate sampling days

As a starting point, sampling days are the same for every river every year. The range of sampling days has to cover the range of days when salmon are abundant in any river. That is determined from -1 sd of abundance distribution for the most early case of river and year, and +1 sd for most delayed case.

```r
# sampling days. Cover expected range across all years and rivers.
j <- which.min(b3); # earliest case
a1 <- b3[j]+background[1];  # value for this case
cat('earliest case',a1, 'timing effect', b1[j],
    'spread effect', b2[j],'\n')
```

```
earliest case 246.9675 timing effect -12.69785 spread effect 1.88978
```

```r
j <- which.max(b4);  # most delayed case
a2 <- b4[j]+background[1];
cat('last case',a2,'timing effect', b1[j],
    'spread effect', b2[j], '\n')
```

```
last case 297.0865 timing effect 9.051173 spread effect 3.02358
```

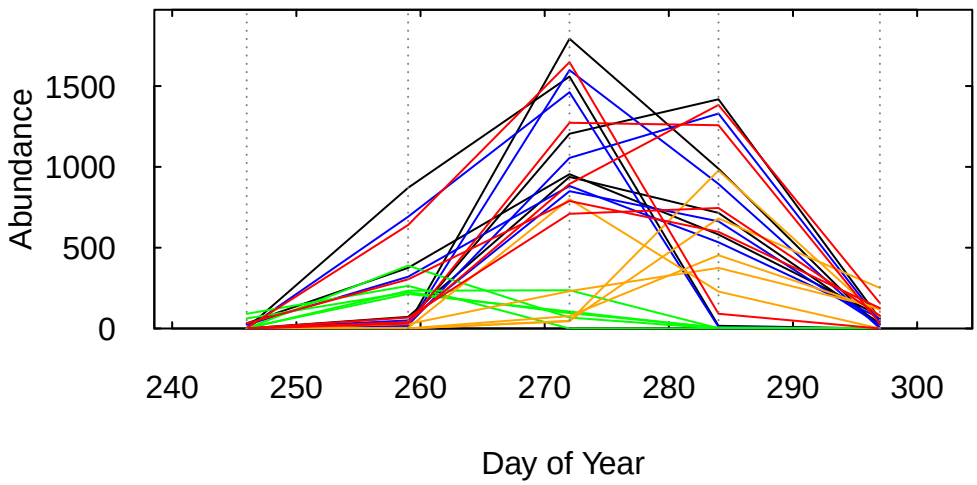
```r
days <- seq(a1, a2, length.out=n_day);
days <- floor(days) # integer days
cat("\n Days of the year for observations:", days,"\n")
```

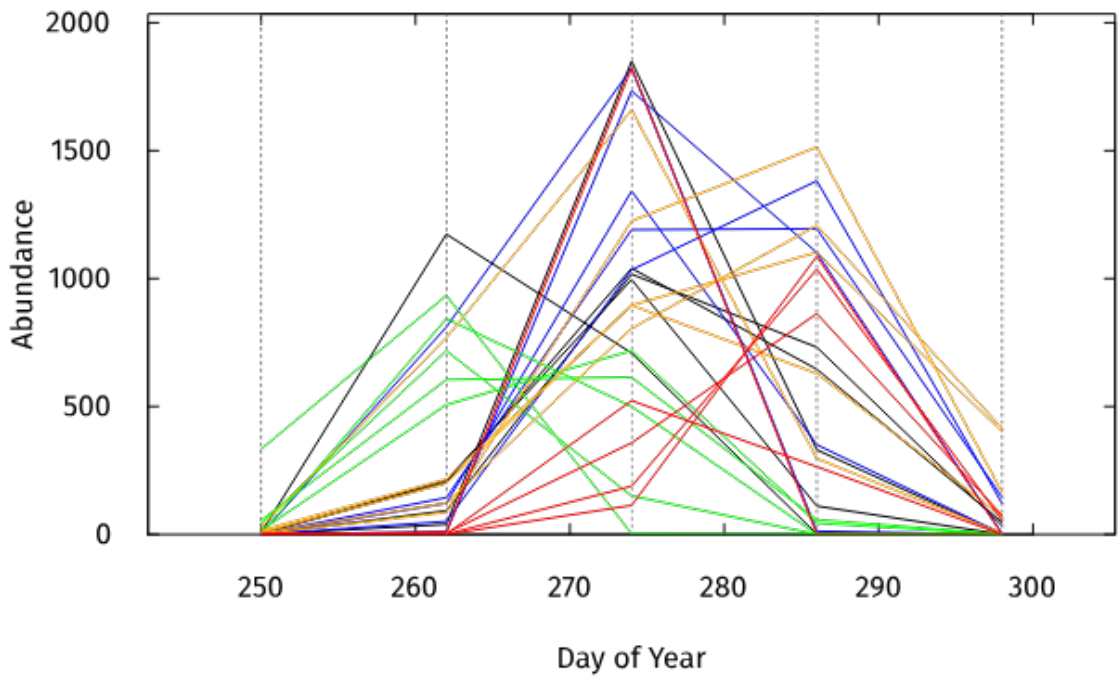
```
 Days of the year for observations: 246 259 272 284 297
```

Abundance by day

The relative abundance of salmon in one river is taken as fixed and independent of years. This is true only when considering the potential abundance of spawners ("capacity" refers to some measure of output of salmon, not input of spawners). The spawning populations in a river may thrive or dwindle for many reasons: fishing, predators, disease,. These random effects within rivers across years will fall into the unexplained variance.

```
#|label: abund_day
# total abundance by year
totalYearly <- rnorm(n_year, 1e5, 0.25e5) # abundance by year
# abundance by day by river by year.
for (river in 1:n_river) {
  for (year in 1:n_year) {
    trueAbundance[,river,year] <- totalYearly[year] * riverEffect[river,3] *
    dnorm(days,
      background[1]+riverEffect[river,1]+ yearEffect[year,1], # mean
      background[2]+riverEffect[river,2]+ yearEffect[year,2]) # SD
  }
}
j <- is.na(trueAbundance) | trueAbundance < 0.1
trueAbundance[j] <- 0

par(tck=0.02, las=1, yaxs="i")
color <- c("black","blue","green","orange","red");
plot(1,1,type="n",
  xlim=c(-5,5)+range(days),
  ylim=c(.9,1.1)*range(trueAbundance),
  xlab="Day of Year", ylab="Abundance");
axis(3,labels=F); axis(4,labels=F);

for (river in 1:n_river) {
  for (year in 1:n_year) {
    lines(days, trueAbundance[,river, year], col=color[river] )
  }
}
abline(v=days, lty="dotted", col="grey50")
```





Area under the curve, Conventional Fit

To establish a contrast to AUC by river by year, we calculate the integral of total salmon abundance from the linear interpolation method and by fitting a normal distribution. Adjust-

ing the linear interpolation by considerations such as a date for unobserved peak spawning is not included (yet).

A spawning salmon may be observed in more than one survey, so a key parameter to translate AUC into salmon abundance is the duration of a spawner in a river. While that duration may be a haphazard estimate, it directly affects abundance estimates and needs to be included in these models. Potentially all of the spawners observed after peak abundance were also observed at peak abundance. This has led to simplifying the survey analyses to be maximum spawners observed. This value is included in the result because the relative abundance of spawners, from some uncalibrated index of abundance, such as maximum spawners observed across multiple surveys in one river and one year, may be sufficient for fisheries management.

Calculations without observation error

Knowing how the precision of estimates of total changes with increasing observation error seems valuable. The baseline is zero observation error. The following code provides functions for exploring the effects of observation error, which I have not yet done. (perhaps Gaussian Process regression?)

function for linear AUC

```
# linear interpolation with irregular intervals.
# each observation applies to half the preceding inteval and half the following.
AUC <- function(days,dat) { # x,y. vectors
  n=length(dat)
  if (length(days) != n) stop("AUC: lengths differ");
  interval <- days[-1] - days[-n]; # second minus first, etc. length n-1
  weight <- interval/2 # n-1 of these
  weight <- c(0,weight)+ c(weight,0); # for non-uniform intervals
  auc <- sum (weight * dat)
#  cat('weight', weight,'\n')
  return(auc)
}
test_dat <- c(1,3,5, 2,1 ); # not symmetrical
test_day <- c(2,5,8,11,14);
AUC(test_day, test_dat ) # 12
```

```
[1] 33
```

function to fit normal distribution

The log of a normal distribution is a parabola

$$log(P_x) = log(\frac{1}{\sigma\sqrt{2\pi}}) + \frac{(x-\mu)^2}{2\sigma^2}$$

$$log(y) = \frac{-1}{2} log(2\pi\sigma^2) + \frac{x^2 - 2\mu x + \mu^2}{2\sigma^2}$$

$$log(y) = -0.5 \left(log(2\pi\sigma^2) + \frac{\mu^2}{\sigma^2} \right) - x\frac{2\mu}{2\sigma^2} + x^2 \frac{1}{2\sigma^2}$$

$$log(y) = a + bx + cx^2$$

where

$a = -.05 \left(log(2\pi\sigma^2) + (\mu/\sigma)^2 \right)$, $b = -0.5\mu/\sigma^2$, and $c = 0.5/\sigma^2$.

After linear regression of $y = a + bx + cx^2$, the mean and standard deviation can be recovered as $\mu = 0.5\ c/b$ and $\sigma = \sqrt{-0.5/c}$.

The Gaussian integral (LaPlace 1782, Gauss 1809) is the *area under the curve* of a normal distribution with $\mu = 0$ and $\sigma^2 = 1/2$.

$$\int_{-\inf}^{\inf} e^{-x^2} dx = \sqrt{\pi}$$

There is an extension applicable to the parabola obtained by regression with the log of abundance data. The antilog of the preceding fitted parabola (parameters a, b, c) is a normal distribution, and there is a analytic solution for the integral of the antilog of a parabola,

$$\int_{-\inf}^{\inf} e^{a+bx-cx^2} dx = \sqrt{\frac{\pi}{c}}\ e^{\frac{b^2}{4c}+a}$$

That integral is the total abundance, *aka* area under the curve, *aka* Gaussian scaling factor. See link.

Fitting to log abundance emphasizes the contribution to the sum of square of residuals from small abundance estimates compared to large. Are small abundance estimates more or less precise than large? Which abundances are more important to estimate accurately? On the other hand, if the precision of each abundance estimate can be quantified, then a weighted

regression can be used. This is deferred, because it is more likely that relative precision is known (ranking, ordinal).

```
#|label: parabola
# the log of a normal distribution is a parabola
AUCnormal <- function(dat, day){
    day2 <- day^2
    y<- log(dat)
    a <- lm (y ~ day + day2)
    return(a)
}
test_dat <- c(1,3,5,2,1); # not symmetrical
test_day <-c(2, 5, 8, 11, 14);
a <- AUCnormal(test_dat, test_day);
p <- coefficients(a)
print(summary(a));
```

```
Call:
lm(formula = y ~ day + day2)

Residuals:
       1        2        3        4        5 
-0.04553  0.01992  0.21339 -0.30445  0.11666 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)  
(Intercept) -1.040916   0.462879  -2.249   0.1535  
day          0.622756   0.135183   4.607   0.0440 *
day2        -0.039767   0.008248  -4.821   0.0404 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2778 on 2 degrees of freedom
Multiple R-squared:  0.9214,	Adjusted R-squared:  0.8429 
F-statistic: 11.73 on 2 and 2 DF,  p-value: 0.07856
```

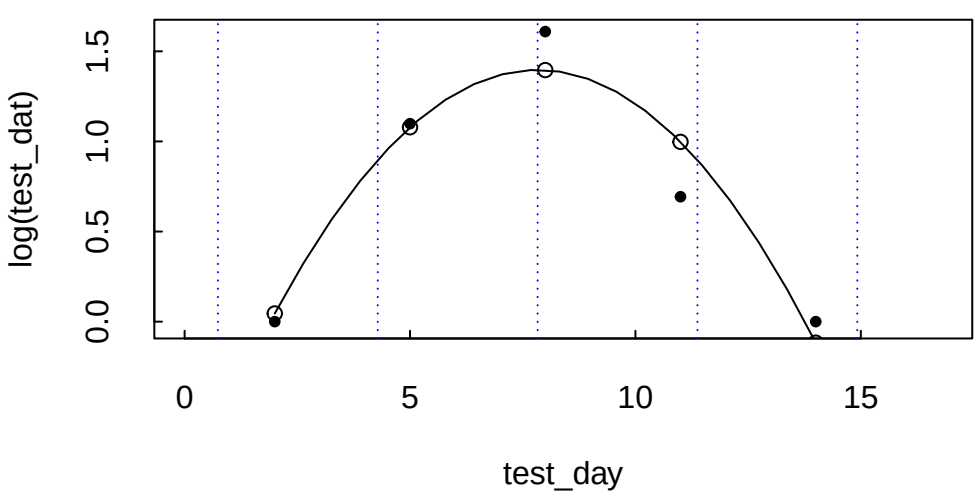
```
print(p);
```

```
(Intercept)         day        day2 
-1.04091605  0.62275564 -0.03976695 
```

```r
y <- predict(a) # data=test_day, default predicted by regression

par(tcl=0.2)
plot(log(test_dat)~test_day, pch=20, yaxs="i",
     xlim=c(0,1.2*max(test_day)),ylim=range(y)*c(.8,1.2));
points(test_day,y)
x= seq(min(test_day), max(test_day), length.out=20)
y <- p[1]+ p[2]*x+p[3]*x^2
lines(x,y)

# extract Gaussian parameters from regression parameter estimates
mu <- -0.5* p[2]/p[3];
sigma <- sqrt(-0.5/p[3]);
# be careful: regression y=a+bx+cx^2 represents math y=a+bx-cx^2
# so see c as -c, see? So,
auc <- sqrt(pi/(-p[3]))*exp( (p[2]^2)/(4*(-p[3])) +p[1]);
cat('\n mu =', mu, 'sigma =', sigma,'auc =',auc, '\n')
```

```
 mu = 7.830066 sigma = 3.545879 auc = 35.94239
```

```r
abline(v= mu+sigma* c(-2,-1,0,1,2), lty='dotted',col='blue')
```

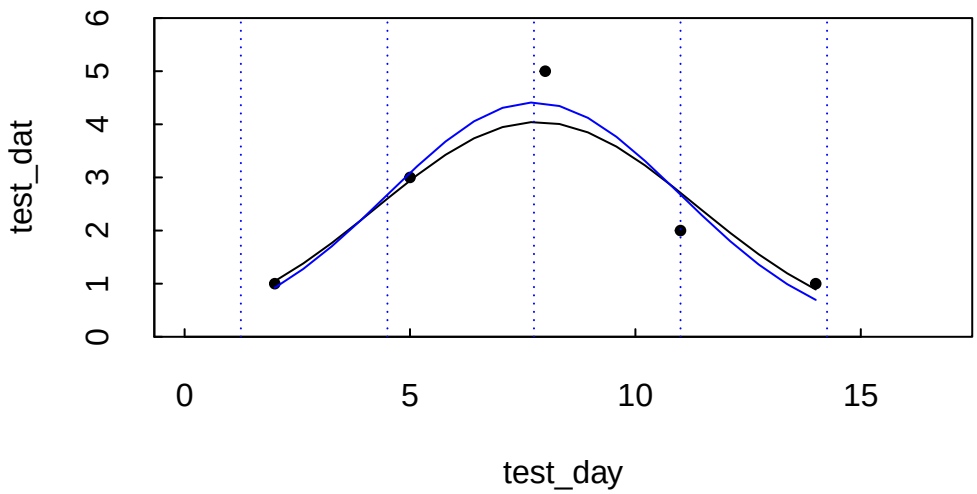



```r
# not logged
plot(test_dat~test_day, pch=20, yaxs="i",
     xlim=c(0,1.2*max(test_day)),ylim=c(0, 1.2*max(test_dat)));
y1 <- exp(y); # antilog
```

```
lines(x, y1);
mean0 = sum(test_dat*test_day) / sum(test_dat);
sd0 = sqrt( sum(test_dat*(test_day-mean0)^2)/(sum(test_dat)-1))
cat('\n simple weighted mean =',mean0,' with sd =', sd0,'\n')
```

```
 simple weighted mean = 7.75  with sd = 3.250874
```

```
y2 <- dnorm(x,mean0,sd0) * auc; # crude normal scaled to total abundance
lines(x,y2,col="blue");
abline(v= mean0+sd0* c(-2,-1,0,1,2), lty='dotted',col='blue')
```



Nonlinear regression (search) for normal distribution

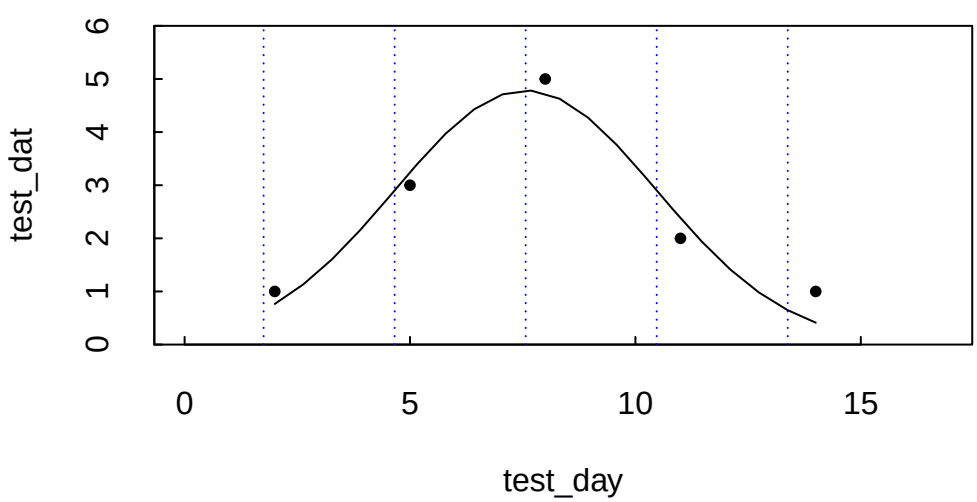
This model minimizes the squares of residuals from trial values of parameters to estimate $\mu, \sigma,$ and N . It has the advantages of (1) fitting a complete distribution to possibly truncated observations (started late or ended early), (2) does not emphasize the importance of small abundances compared to large, and (3) provides a direct estimate of the precision of the fitted value for AUC.

```
#|label: fitGauss
# the model and the optimization in one function
fitGaussian <-function(x,y,mu,sigma,scale){
  f = function(par){  # par is c(mu,sigma,scale), searched by optim()
    d = par[3]*dnorm(x,mean=par[1],sd=par[2]) # predicted from trial values in par
    ssq=sum((d-y)^2) # minimize SSQ (or another distance)
```

```
    return(ssq)
  }
  optim(par=c(mu,sigma,scale),f, hessian=T)
  # search for minimum, starting from guesses.
  # Return curvature of SSD manifold at best fit for parameters (Hessian)
}
a <- fitGaussian(test_day, test_dat, mu=7.8,sigma=3.5, scale=36);
mean1 <- a$par[1];
sd1 <- a$par[2];
auc1 <- a$par[3];
SSres <- a$value;
se <- sqrt(diag(solve(a$hessian)));
r2 <- 1-SSres/sum((test_dat - mean(test_dat))^2); # SSreg/SSmean

cat(sep='', 'mean=',  round(mean1,2), '(', round(se[1],2),
    '), sd=',  round(sd1,2),   '(', round(se[2],2),
    '), auc=', round(auc1,2),  '(', round(se[3],2),
    '), r2=',  round(100*r2,0),'% \n' );
```

```
mean=7.57(0.46), sd=2.91(0.56), auc=34.86(5.17), r2=94%
```

```
par(tcl=0.2);
plot(test_dat~test_day, pch=20, yaxs="i",
     xlim=c(0,1.2*max(test_day)),ylim=c(0, 1.2*max(test_dat)));
lines(x, auc1*dnorm(x, mean1, sd1))
abline(v= mean1+sd1* c(-2,-1,0,1,2), lty='dotted',col='blue')
```



Monte Carlo fit

As the first step to a Bayes-LaPlace approach to fitting a multi-level regression, the conventional one river in one year approach with Hamiltonian Monte Carlo. This uses the Stan language for Bayesian statistics via R package *rstan*.

Over-fitting is a problem for prediction in ecological systems with a superabundance of habitat indicators. Leave-one-out statistics, where the predicted value of each data point, when it is omitted from the data used to fit the model, is a safeguard against over-fitting. The latest development [*as of writing*] of this is "PSIS-LOO" (Betancourt *et al.*, 2017), available via R package *loo*.

experience: scheduling, downloading, booting R_stan took ~70 seconds. The following test of rstan completed in 76 seconds, mainly compiling. Sampling took less than 1/10 second.

```
#|label: test-rstan
#|eval: false
#|echo: false
# library(rstan)
stancode <- '
    data {real y_mean;}
    parameters {real y;}
    model {y ~ normal(y_mean,1);}' # from  R help: stan_model
 mod <- stan_model(model_code = stancode)
```

```
Trying to compile a simple C file


Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c

   File /var/folders/qk/mzcctd911wbbk73jwtfz8hf40000gn/T//Rtmp5q2a1E/callr-uev-1261b1e5128ef
      my_client <- gargle::gargle_oauth_client(
      )
   They were ignored

using C compiler: ‘Apple clang version 14.0.0 (clang-1400.0.29.202)’
using SDK: ‘’
clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Libra
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/S
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
/Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
```

```
         ^~~~~~~
1 error generated.
make: *** [foo.o] Error 1
```

```r
fit <- sampling(mod, data = list(y_mean = 0))
```

```
SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 3.1e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.31 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.071 seconds (Warm-up)
Chain 1:                0.12 seconds (Sampling)
Chain 1:                0.191 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 2.8e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.28 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
```

```
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.149 seconds (Warm-up)
Chain 2:                0.279 seconds (Sampling)
Chain 2:                0.428 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 7e-06 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.07 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.091 seconds (Warm-up)
Chain 3:                0.073 seconds (Sampling)
Chain 3:                0.164 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 1.8e-05 seconds
```

```
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.18 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.044 seconds (Warm-up)
Chain 4:                0.044 seconds (Sampling)
Chain 4:                0.088 seconds (Total)
Chain 4:
```

```r
print(fit)
```

```
Inference for Stan model: anon_model.
4 chains, each with iter=2000; warmup=1000; thin=1;
post-warmup draws per chain=1000, total post-warmup draws=4000.

      mean se_mean   sd  2.5%   25%   50%   75% 97.5% n_eff Rhat
y     0.02    0.02 1.00 -1.94 -0.68  0.00  0.69  2.02  1647    1
lp__ -0.50    0.02 0.73 -2.57 -0.64 -0.23 -0.05  0.00  1775    1

Samples were drawn using NUTS(diag_e) at Tue Aug  6 13:46:06 2024.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).
```

```r
fit2 <- sampling(mod, data = list(y_mean = 5))
```

```
SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
```

```
Chain 1:
Chain 1: Gradient evaluation took 6e-06 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.06 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.033 seconds (Warm-up)
Chain 1:                0.041 seconds (Sampling)
Chain 1:                0.074 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 7e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.07 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
```

```
Chain 2:
Chain 2:  Elapsed Time: 0.05 seconds (Warm-up)
Chain 2:                0.073 seconds (Sampling)
Chain 2:                0.123 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 1.7e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.17 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.085 seconds (Warm-up)
Chain 3:                0.157 seconds (Sampling)
Chain 3:                0.242 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 1e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.1 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
```

```
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.11 seconds (Warm-up)
Chain 4:                0.124 seconds (Sampling)
Chain 4:                0.234 seconds (Total)
Chain 4:
```

```r
print(fit2)
```

```
Inference for Stan model: anon_model.
4 chains, each with iter=2000; warmup=1000; thin=1;
post-warmup draws per chain=1000, total post-warmup draws=4000.

      mean se_mean   sd  2.5%   25%   50%   75% 97.5% n_eff Rhat
y     4.98    0.02 0.99  3.05  4.31  4.95  5.61  6.92  1724    1
lp__ -0.49    0.02 0.72 -2.37 -0.64 -0.22 -0.05  0.00  1610    1

Samples were drawn using NUTS(diag_e) at Tue Aug  6 13:46:11 2024.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).
```

```r
#  ## fit
# 4 chains, each with iter=2000; warmup=1000; thin=1;
# post-warmup draws per chain=1000, total post-warmup draws=4000.
#       mean se_mean   sd  2.5%   25%   50%   75% 97.5% n_eff Rhat
# y     0.02    0.03 1.00 -1.89 -0.65  0.03  0.68  2.01  1473    1
# lp__ -0.50    0.02 0.71 -2.54 -0.66 -0.22 -0.05  0.00  1528    1
#
#  ## fit2
# 4 chains, each with iter=2000; warmup=1000; thin=1;
# post-warmup draws per chain=1000, total post-warmup draws=4000.
#       mean se_mean   sd  2.5%   25%   50%   75% 97.5% n_eff Rhat
# y     4.99    0.03 0.99  3.01  4.32  4.97  5.68  6.95  1297    1
# lp__ -0.49    0.02 0.69 -2.55 -0.64 -0.23 -0.05  0.00  1811    1
```

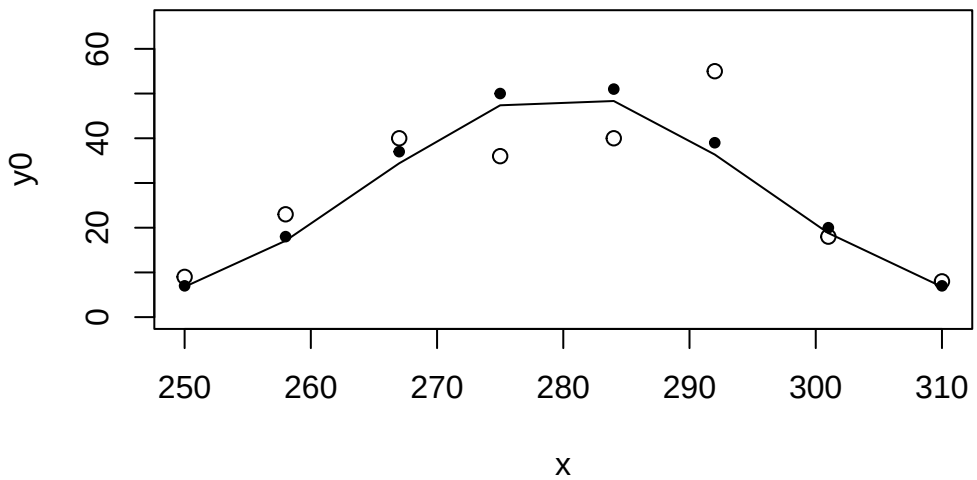
```r
#|label: stan_Gauss
#|cache: true
# library(rstan)
stan_Gauss_code <- "
functions {
    // Gaussian probability density distribution
    // faster than (1/(sigma*sqrt(2pi)))*exp(-(1/2)*((x-mu)/sigma)^2)
    //  1/sqrt(2*pi) = 0.3989423
  vector Gauss(vector x, real mu, real sigma) {
    real inv_sigma;
    inv_sigma = 1./sigma;
    return 0.3989423*inv_sigma*exp(-0.5* square((x-mu)*inv_sigma));
  }
}
data {
  int<lower=0> np;  // count of prior (8)
  vector[np] prior; // timing: m,s; spread: m,s; auc: m,s
  int<lower=0> n;   // count of data
  vector[n] x;      // day of year for observation
  vector[n] y;      // estimate of abundance for that day
}
parameters {
  real <lower=0> timing;  // mean of Gaussian
  real <lower=0> spread;  // standard deviation of Gaussian
  real <lower=0> auc;     // area under curve, total abundance
  real <lower=0> sigma;   // sd of residuals
}
model {
  vector[n] ypred;
  // prior knowledge
  timing ~ normal(prior[1], prior[2]);
  spread ~ normal(prior[3], prior[4]);
  timing ~ normal(prior[5], prior[6]);
  sigma ~ gamma( prior[7], prior[8]);
  // prediction from trial parameters
  ypred = auc * Gauss(x, timing, spread);  // Gauss returns a vector
  // likelihood of data given prediction
  y~normal(ypred, sigma);
} "

stanc_ret <- stanc(model_code = stan_Gauss_code) # check syntax, compile
stan_Gauss <- stan_model (stanc_ret=stanc_ret) # create model
```

```
Trying to compile a simple C file


Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c

   File /var/folders/qk/mzcctd911wbbk73jwtfz8hf40000gn/T//Rtmp5q2a1E/callr-uev-1261b26bb1af3
      my_client <- gargle::gargle_oauth_client(
      )
   They were ignored

using C compiler: ‘Apple clang version 14.0.0 (clang-1400.0.29.202)’
using SDK: ‘’
clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Libra
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/S
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
/Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
         ^~~~~~~
1 error generated.
make: *** [foo.o] Error 1
```

```r
#|label: test_and_guess
# create test data and initial guesses
# background temporal distribution
timing <- 280.0; # day of year mean (peak) spawner abund.
spread <-  15.0; # std dev of Gaussian spawner abund.
auc <- 2000.0;   # abundance of spawners
n = 8; # number of obsevations, number of days
# x is day of observation (day of Julian year)
x <-  seq(from=timing-2*spread,to=timing+2*spread, length.out=n) %>% floor;
# y is estimates of abundance
y0 <- (auc * dnorm(x,timing, spread)) %>% round(0); # without error
y  <- (y0 *  rnorm(y0,1,0.2))         %>% round(0); # 20% error
cat(' x  =',x,'\n y0 =',y0,'\n y  =',y,'\n');
```

```
 x  = 250 258 267 275 284 292 301 310
 y0 = 7 18 37 50 51 39 20 7
 y  = 9 23 40 36 40 55 18 8
```

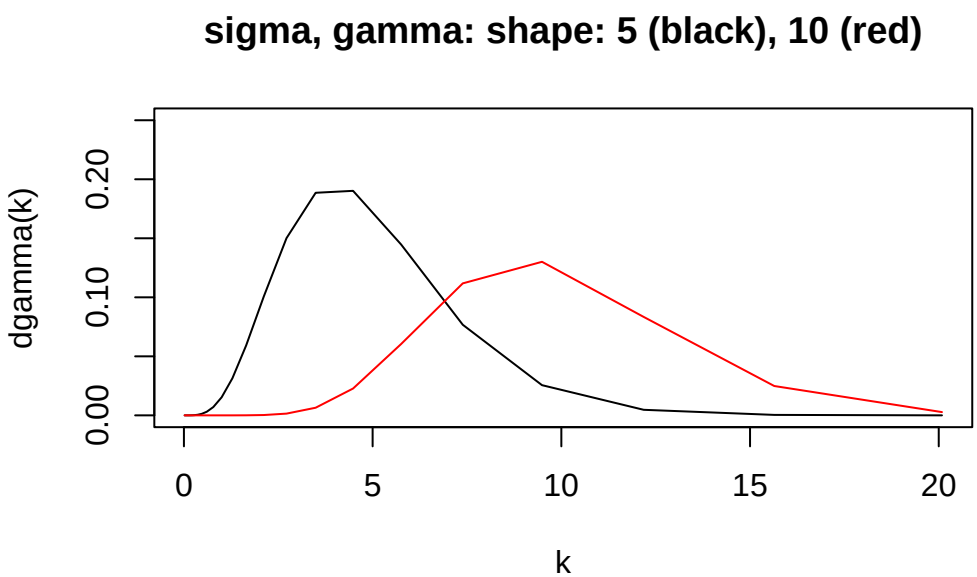
```r
# ----- priors ------
    # guess for auc is linear AUC
auc <- round(AUC(x,y), 0);
    # guess for timing is day weighted by count
ptm <- round(sum(x*y)/sum(y), 0); # ptm abbreviates prior timing mean
    # guess for spread is weighted variance
psm <-  round(sqrt(sum( y* (x-ptm)^2)/sum(y)), 0) # ignore supposed bias

plot(x, y0, pch=20, ylim=c(0,1.2*max(y)) );
    points(x,y); lines(x,auc*dnorm(x,ptm,psm));
```




```r
cat(sep='', 'calculated priors: timing =',ptm,', spread = ',psm,', AUC = ',auc,'\n')
```

```
calculated priors: timing =280, spread = 15, AUC = 1883
```

```r
# sigma (sd of normal residuals) has prior gamma(shape=5, scale=1)
k=exp(seq(-4,3,.25));  # interested in behaviour near zero

plot( k,dgamma(k,shape=5, rate=1), type="l",ylim=c(0,0.25),
      ylab='dgamma(k)', main='sigma, gamma: shape: 5 (black), 10 (red)' );
lines(k,dgamma(k,shape=10,rate=1), col="red");
```




```r
prior <- c(
  timing_m = ptm,       # 1. about Julian day 280 (Sept. 30 is 273)
  timing_s = 7.0,       # 2. range, 4 sd is 1 moon orbit
  spread_m = psm,       # 3. about 2 moons.
  spread_s = 3,         # 4. range 1 to 4 moons. Avoid negative!
  auc_m    = auc,       # 5. simplest estimate
  auc_s    = 0.25*auc,  # 6. expect 0/5 to 1/5 of estimate
  sigma_shape = 10.0,   # 7. high and wide. Y should be scaled: y/max(y)
  sigma_scale = 1.0)    # 8.
print(prior) # shows names
```

```
   timing_m    timing_s    spread_m    spread_s       auc_m       auc_s
     280.00        7.00       15.00        3.00     1883.00      470.75
sigma_shape sigma_scale
      10.00        1.00
```

```r
dat <- list(np=8, prior=prior, n=8, x=x, y=y);

# might need to initialize HMC
# one chain for testing
# init1 <- list(
#  chain1=list(timing=ptm, spread=psm, auc=auc, sigma=10.0));
# print(as.data.frame(init1));
#  four chains for fitting
# init4 <- list(
```

```
#  chain1=list(timing=ptm*1.10, spread=psm*1.10, auc=auc*1.10, sigma=12.0),
#  chain2=list(timing=ptm,1.05, spread=psm*1.05, auc=auc*1.05, sigma=10.0),
#  chain3=list(timing=ptm*0.95, spread=psm*0.95, auc=auc*0.95, sigma=8.0),
#  chain4=list(timing=ptm*0.90, spread=psm*0.90, auc=auc*0.90, sigma=6.));
# print(as.data.frame(init4));

# I used these print statements in model block to debug Stan code
#  with (iter=1, chains=1);
#      // print("timing =",timing); // to check Stan code
#      // print("spread =",spread);
#      // print("auc =",auc);
#      // print("sigma =",sigma);
#      // print("x =",x);
#      // print("y =",y);
#      // print("ypred =",ypred);
```

Test function Gauss()

test my function in Stan function block by using expose_stan_functions

```
#|label: test_Gauss
#|cache: true
expose_stan_functions(stan_Gauss);
```

```
Trying to compile a simple C file


Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c

   File /var/folders/qk/mzcctd911wbbk73jwtfz8hf40000gn/T//RtmposvH0t/callr-uev-129ff7ace9732
      my_client <- gargle::gargle_oauth_client(
      )
   They were ignored

using C compiler: ‘Apple clang version 14.0.0 (clang-1400.0.29.202)’
using SDK: ‘’
clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Libra
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/S
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
```

```
/Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
         ^~~~~~~
1 error generated.
make: *** [foo.o] Error 1
```

```r
Gauss(c(2,3,4), 0, 1);
```

```
[1] 0.0539909692 0.0044318486 0.0001338302
```

```r
  # [1] 0.0539909665 0.0044318484 0.0001338302
 dnorm(c(2,3,4), 0,1)
```

```
[1] 0.0539909665 0.0044318484 0.0001338302
```

```r
  # [1] 0.0539909665 0.0044318484 0.0001338302
```

Fit stan_Gauss

```r
#|label: fit_stan_Gauss
#|warning: false
#|cache: true
fit <- sampling(stan_Gauss, data=dat, chains=4, iter=2000); # init=init1);
```

```
SAMPLING FOR MODEL 'file1261b338ae9d4' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 5.7e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.57 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
```

```
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.093 seconds (Warm-up)
Chain 1:                0.068 seconds (Sampling)
Chain 1:                0.161 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'file1261b338ae9d4' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 1.5e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.15 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.08 seconds (Warm-up)
Chain 2:                0.079 seconds (Sampling)
Chain 2:                0.159 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'file1261b338ae9d4' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 1.5e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.15 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
```

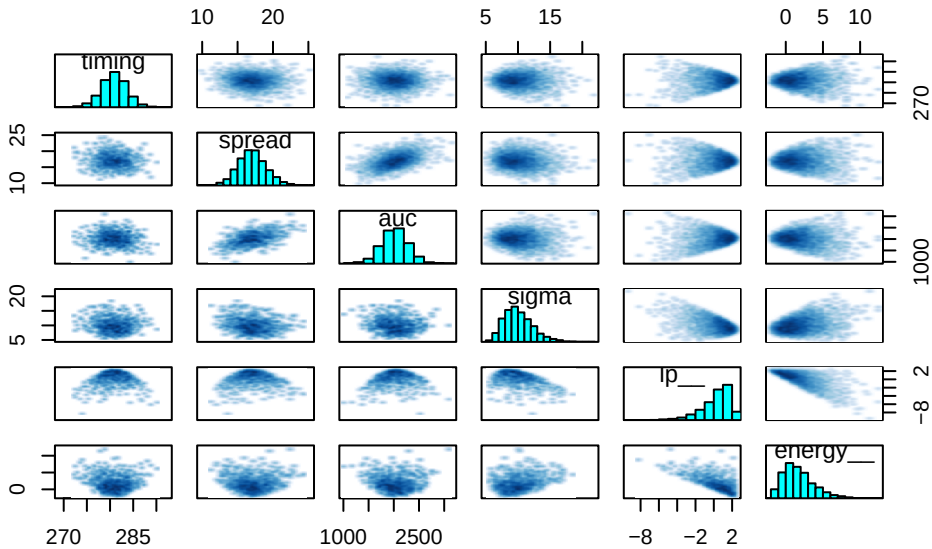
```
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.096 seconds (Warm-up)
Chain 3:                0.045 seconds (Sampling)
Chain 3:                0.141 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'file1261b338ae9d4' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 3.4e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.34 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.103 seconds (Warm-up)
Chain 4:                0.066 seconds (Sampling)
Chain 4:                0.169 seconds (Total)
Chain 4:
```

```r
print(fit)
```

```
Inference for Stan model: file1261b338ae9d4.
4 chains, each with iter=2000; warmup=1000; thin=1;
post-warmup draws per chain=1000, total post-warmup draws=4000.

          mean se_mean     sd    2.5%     25%     50%     75%   97.5% n_eff
timing  280.88    0.05   2.76  275.13  279.13  280.93  282.69  286.14  3709
spread   17.07    0.03   2.01   13.28   15.73   17.01   18.30   21.26  3427
auc    2021.68    4.90 269.28 1482.20 1850.83 2021.53 2193.40 2547.16  3019
sigma    10.07    0.04   2.23    6.52    8.43    9.78   11.42   15.28  2984
lp__      0.27    0.04   1.56   -3.64   -0.51    0.63    1.42    2.20  1599
       Rhat
timing    1
spread    1
auc       1
sigma     1
lp__      1

Samples were drawn using NUTS(diag_e) at Tue Aug  6 13:48:52 2024.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).
```

```r
pairs(fit, condition=0.2); # plot
```

```
Warning in par(usr): argument 1 does not name a graphical parameter
Warning in par(usr): argument 1 does not name a graphical parameter
Warning in par(usr): argument 1 does not name a graphical parameter
Warning in par(usr): argument 1 does not name a graphical parameter
Warning in par(usr): argument 1 does not name a graphical parameter
Warning in par(usr): argument 1 does not name a graphical parameter
```



Experience

Stan is *strongly typed* to enforce rigor in coding, an attempt to avoid subtle bugs. Which takes some getting used to, a contrast to R. I found **vector[n] x** confusing compared to **real x[n]**, the latter being an array, so you could have **real[m] x[n]** which would be n arrays of m values. And then row_vectors which are not vectors or arrays, nor rows or columns of a matrix. I see the point: protect domain experts from their otherwise sloppy linear algebra.

This model does not work with the default improper priors, the sampling heads off to spread $\rightarrow \infty$ despite initializing in the vicinity of the best fit. The prior knowledge assigned to parameters is derived from simple models applied loosely (large values for the prior regarding range of a parameter). Because there is adequate data, the effect of prior knowledge will be small. When there is a lot of data, by modeling many years and rivers at one, the effect of prior knowledge (*a priori* probability distributions for parameters) on the results (ditto but *a posteriori*) will be smaller again.

Parameters were recovered accurately, but that is because the true model was known. The true model is unknown (and effectively infinitely complicated) and a serviceable while simplistic description will be discovered from exploring real data.

Many Rivers, Many Years

Multilevel Regression

A simple, linear, multivariate, multilevel regression model would be

$$
\begin{aligned}
y &= \mathcal{N}(\hat{y},\ \sigma) \\
&= \mathcal{N}(\alpha + X\beta + Z\gamma,\ \sigma)
\end{aligned}
$$

where $\boldsymbol{X}$ is the matrix of independent variables and the fitted parameters α (intercepts) and β (slopes) define the overall or average linear regression, the "fixed effects". If there were no more to the model, this would describe 'complete pooling" of the data.

The matrix $\boldsymbol{Z}$ describes the groups in $\boldsymbol{X}$ and γ is the fitted estimate of the deviation of the parameter α by groups (easily extended to have β also vary by group). Thus $Z\gamma$ makes this similar to a *random effects* model, where, typically, the intention is to discern the fixed effects and discard the random effects: what is behind the noise that can be accounted for via groups?

In the Bayes-Laplace paradigm, the group effects are real, important for prediction, and modeled as random draws from a fitted distribution. Thus the group effects are not independent: information is shared. The choice of the group effects distribution involves prior knowledge (thus *hyper-priors* as pointless jargon) of its parameters. Those parameters are fitted as part of this model. The observed (dependent) variable to be predicted by the model, y, and the residuals (predictions minus observations) are assumed (more prior knowledge) to have a Gaussian (normal) distribution $\mathcal{N}$ with standard deviation σ.

Some quotes, about how "partial pooling" shares information among groups, might be helpful at this point:

> "An extreme approach would be to completely pool all the data and estimate a common vector of regression coefficients . At the other extreme, an approach with no pooling assigns each group its own coefficient vectorβ that is estimated separately from the other groups. A hierarchical model is an intermediate solution where the degree of pooling is determined by the data and a prior on the amount of pooling." - Stan User's Guide version 2.19 page 23.

> " 'While complete pooling or no pooling of data across groups is sometimes called for, models that ignore the grouping structures in the data tend to underfit or overfit' (Gelman *et al.*, 2013). Hierarchical [multi-level] modeling provides a compromise by allowing parameters to vary by group at lower levels of the hierarchy while estimating common parameters at higher levels. **Inference for each group-level parameter is informed not only by the group-specific information contained in the data but also by the data for other groups as well.** This is commonly referred to as *borrowing strength* or *shrinkage*."
> - http://mc-stan.org/rstanarm/articles/glmer.html

River and Year Effects on Timing and Spread

The background model for temporal distribution of salmon spawner abundance is two parameters, timing and spread. This is applied to all years and all rivers. The other part of the background is the overall abundance of salmon for each year, the run size. If the data for all rivers were pooled, there would only be $n_year + 2$ parameters.

The departure from that background for any one river, the river effect on timing and spread, is two parameters. In this model, the prior knowledge for the probability distributions for the river effect parameters proposes they are zero unless the data unambiguously supports the existence of such a year effect (across all of the rivers). The exponential probability distribution provides that effect, $Pr(x) = \beta e^{-\beta x}$. This idea is used for lasso regression where the objective is to remove or ignore parameters that are unimportant.

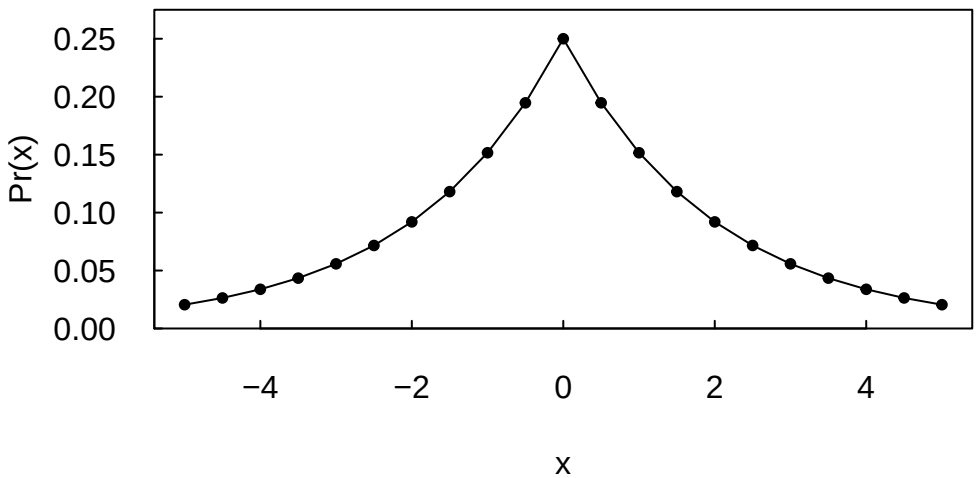
To increase the effect of priors for parameters where the posterior marginal mean value (fitted value) is likely near zero, I used the LaPlace function, a double exponential. This was applied to the fitted distribution for year effects and river effects, because these are departures from an independently fitted overall mean for years and rivers, so year_timing, year_spread, river_timing, and river_spread have LaPlace$(0, \sigma)$ priors (Stan Functions Reference Version 2.32, section 19.7 page 206).

Example of double exponential PDD, the LaPlace distribution

```
#|label: double_exp
# example of double exponential probability density distribution
LaPlace <- function(x, mu, sigma){
    # // (1/(2*sigma)) * exp(-abs(x-mu)/sigma)
    inv_sigma= 1.0/sigma
    result = 0.5 * inv_sigma * exp(-abs(x-mu) * inv_sigma)
    return(result)
}
x <- seq(-5, 5, 0.5); # peak will be at sigma/2
mu <- 0.
sigma <- 2.
y <- LaPlace(x, mu, sigma);
cat('sum y = ', sum(y), '\n')
```

```
sum y =  1.865903
```

```
par(las=1,tcl=0.2, pch=20, yaxs="i");
plot(y~x, type='o', ylim=c(0.0, 1.1/(2*sigma)), ylab='Pr(x)' );
```



River effects are modeled the same way. These are fixed effects. It is likely a population in specific river will have a trend or a discontinuity, that could be a helpful parameter and might be related to a habitat variable for that river (flows, temperature, events). Correlations of timing and/or spread between rivers are likely (rivers behaving the same in terms of spawners), and across years (trends, shifts). There is no *year* × *river* effect, the ideosyncratic departure of a river in a year from expected norms is relegated to the unexplained variance.

- Conceptually, if spawner timing for some year (mean day over all rivers) is late, then it may have less spread as a result of waiting for a migration barrier to drop.
- Conversely, if spawner timing for some river (mean day over all years) is late it may have more spread perhaps because of an lengthy and arduous up-river migration (leading to diffusion of timing of arrival).

Those example indicate there will be habitat indicators that are covariates (predictors) for the stream and year effects on temporal distributions. I look forward to this model reapplied as the foundation for such extensions.

River and Year Effects on Abundance

The area under the curve of spawner abundance is not the same as the number of spawners because a spawner will reside in the spawning ground for many days. That **residence time** may be longer for females guarding redds than for males that drift down-river after spawning. A model that involved the accumulation of identified spawners would involve a cumulative distribution function, possibly preferable. But in this case the total annual abundance of spawners over all rivers in a year, is not the run size considered as the number of salmon independent of multiple observations of the same fish. Determining number of fish from the sum of daily abundances of fish requires knowing the duration of a fish in a river. That will

be an important calibration parameter, but probably complicated by the need for habitat indicators: perhaps warmer river temperatures mean shorter spawning ground residence.

Abundance is primarily a **year effect**. There is little prior knowledge other than *non-negative*, in this model, but that will not be typical. Autocorrelations apply, such as known long trends in salmon abundance, and four-year generation cycles (returns are largely age 1.2 (one freshwater year after spawning previous fall, 2 marine years), dependent on spawners four years previous). The application of this model is likely to involve known abundances for all but the last year.

Independent of years (or average over all years) there is a **river effect** for abundance: the proportion of the total that goes to each river. Perhaps proportions vary due to spawning ground area and/or quality, or river length. The existence of sub-populations (or populations within a metapopulation, a conservation unit, CU) of salmon that home to one stream and not to others is an important complication. This model assumes that any such sub-populations vary identically among years. The proportions by river sum to the entire abundance of salmon, therefore the fitted values for that vector of parameters (with length *n_river*) are not independent, they must sum to unity. Such a vector is a simplex and a specific probability model is required to meet that constraint, the Dirichlet distribution (Gelman et al 2017, page).

After considering temporal distributions and abundance by year and river proportion, the abundance in each year in each river (i.e. the AUC) is estimated as

$$\hat{y}_{i,j,k} = \mathcal{N}(\text{day}_i, \mu_{j,k}, \sigma_{j,k})$$

where the timing $\mu_{j,k}$ for river j and year k involves $n_river \times n_year$ parameters. We might expect those parameters to have spatial correlation length scales (near or distant in measures of river properties) and temporal correlations (successive generations, population trends,) thus scope for additional data and knowledge to sharpen this type of model.

The daily observations are predicted from all of these parameters. The probability of that prediction, given the data and an assumption about the distribution of residuals, is the basis for describing the joint probability density in the parameter space, and thereby estimates of the parameters and their marginal distributions, e.g. median (50% quantile) with 10% and 90% quantiles to indicate precision. While the correlations between parameters are typically ignored for reporting, they is required for predicting. One aspect of predicting is to simulate data described by the fitted model and see if that looks like the observed data.

River proportion

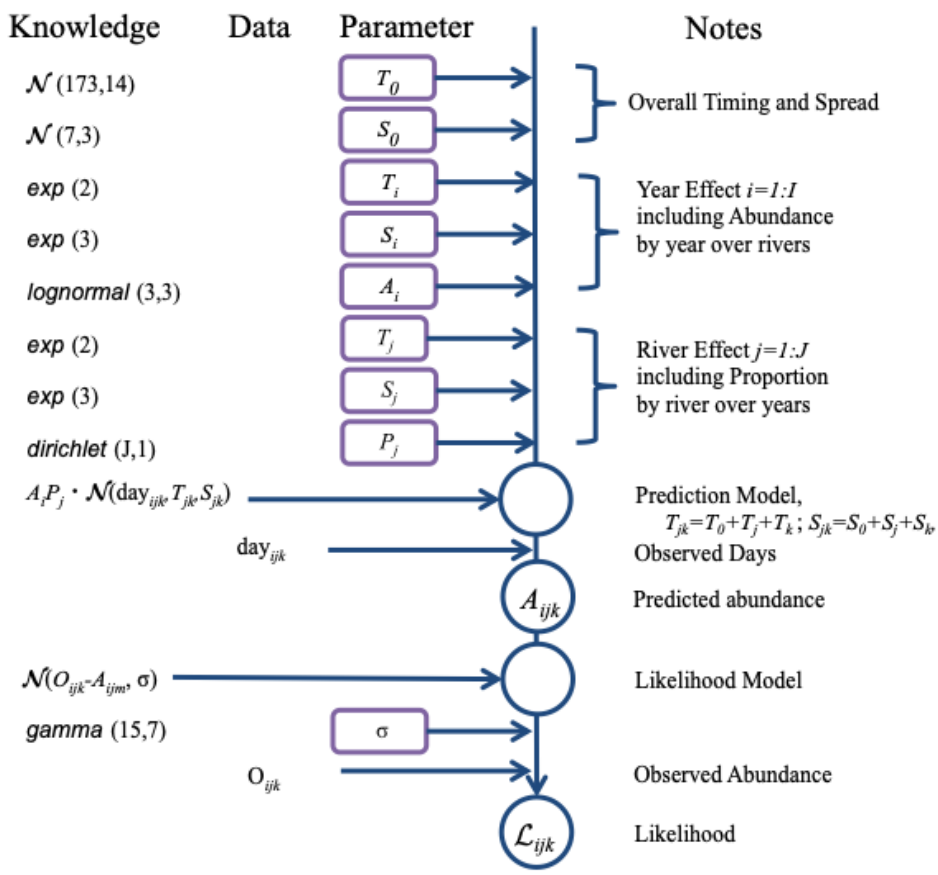
Describing the relative abundance of spawners between rivers as the proportion of salmon that enter that river means the fitted estimates for those proportions across all river must sum to 1. That vector of non-negative parameters is a *unit simplex*, the Direchlet distribution (Stan Functions Reference v2.19 §23.1 p126) describes the *multinomial* probabilities for values

within a simplex. The Direchlet parameter describes how even the probabilities are, with large values (>100) creating a even distribution and small values (<1/100) creating extremes (one of ~1 and all others ~zero). Using 1 as the default looks like uniform random.

```
data {
int <lower=1> n_river;
real <lower=0> alpha; // large alpha means even proportions.
}
parameters {
simplex [n_river] river_prop; // constrained 0<x<1 and sum=1.
}
model {
river_prop ~ direchlet(alpha); // sampling for run proportions
}
```

Model Overview

> "In theory, a Bayesian model should include all relevant substantive knowledge and subsume all possible theories. In practice, it won't."
> - Jeffrey B. Arnold
> https://jrnold.github.io/bayesian_notes/model-checking.html



Any assumed model is one of many possible variants, so we need a basis for comparing models. How the model fits data can be based on prediction ability using *cross validation* that will *leave one out* (LOO) when fitting, and then predict that *left out observation* from the fit. Conceptually, the fitting is repeated for every data point, but there is a mathematical alternative. The required information is the log likelihood of each data point; produced and saved in the *generated quantities* block in a Stan model. See http://mc-stan.org/loo/reference/loo-package.html.

Stan code

This code uses an overly simplified version of salmon abundance patterns, but tries to get the surrounding stats model right.

"Variables declared in the model block are always local to the model block and may not be accessed in the generated quantities block; to make a variable accessible in the model and generated quantities block, it must be declared as a transformed parameter."

Present model postpones this block, deferring LOO and number of *effective parameters.*

```
#|label: multi_gauss
#|cache: true
stan_MultiGauss_code <- '
functions {
    // Gaussian probability density distribution
    // faster than (1/(sigma*sqrt(2pi)))*exp(-(1/2)*((x-mu)/sigma)^2)
    //  1/sqrt(2*pi) = 0.3989423
//vector Gauss(vector x, real mu, real sigma) {  // vector version
  real Gauss(real x, real mu, real sigma) {      // scalar version
    real inv_sigma;
    inv_sigma = 1./sigma;
    return 0.3989423*inv_sigma*exp(-0.5* square((x-mu)*inv_sigma));
  }
}
data {
  int<lower=0> n_prior;     // length of vector of priors
  vector[n_prior] prior;    // background, rivers, years
  int<lower=0> n_year;      // how many years
  int<lower=0> n_river;     // how many rivers
  int<lower=0> n_data;      // how many data: all days, rivers, years
  vector[n_data] day_obs;   // day of year for each observation
  vector[n_data] y_obs;     // estimate of abundance for each day in x
  int year_index  [n_data]; // id  of year in data, integer
  int river_index [n_data]; // id of river in data, integer
  vector [n_river] river_prop_prior; // Dirichelet, vector of ones
}
parameters {
    real <lower=0> timing;         // all years and rivers
    real <lower=0> spread;         // all years and rivers
    vector <lower=0>[n_year] run;  // annual spawner abundance
    vector[n_year]   year_timing;  // by year, all rivers
    vector[n_year]   year_spread;  // by year, all rivers
    vector[n_river]  river_timing; // by rivers, all years
    vector[n_river]  river_spread; // by rivers, all years
    simplex[n_river] river_prop;   // fraction of run
    real <lower=0> sigma;          // stdev of Normal residuals
}
model {
// local variables
  vector[n_data] y_pred; // predicted
  int year;              // index
  int river;             // index
```

```
  real timing_pred;      // background+yearEffect+riverEffect
  real spread_pred;      // background+yearEffect+riverEffect
// prior knowledge of prob. distributions of parameters
  timing            ~ normal(prior[1], prior[2]);
  spread            ~ normal(prior[3], prior[4]);
  run               ~ lognormal(prior[5], prior[6] );
  year_timing       ~ normal(0,prior[7]);  // effect not result
  year_spread       ~ normal(0,prior[8]);  // effect not result
  river_timing      ~ normal(0,prior[9]);  // effect not result
  river_spread      ~ normal(0,prior[10]); // effect not result
  river_prop        ~ dirichlet(river_prop_prior);
  sigma             ~ gamma(prior[11], prior[12]);
// predict from a sampled point in parameters space
  for (j in 1:n_data){
    year          = year_index[j];
    river         = river_index[j];
    timing_pred   = timing + year_timing[year] + river_timing[river];
    spread_pred   = spread + year_spread[year] + river_spread[river];
    y_pred[j]      = run[year] * river_prop[river] *
                     Gauss(day_obs[j],timing_pred,spread_pred);
  }
// accumulate log likelihood of data given prediction
  y_obs ~ normal(y_pred, sigma); // absolute or relative error?
}
// generated quantities { // postponed
//    vector[n_data] log_lik;
//    log_lik = normal_lpdf(y_pred | y_obs, sigma);
//} '

stanc_ret <- stanc(model_code=stan_MultiGauss_code) # check fussy syntax, compile
stan_MultiGauss <- stan_model (stanc_ret=stanc_ret) # create model
```

```
Trying to compile a simple C file

Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c

   File /var/folders/qk/mzcctd911wbbk73jwtfz8hf40000gn/T//Rtmp5q2a1E/callr-uev-1261b3ca5cdc9
      my_client <- gargle::gargle_oauth_client(
      )
   They were ignored
```

```
using C compiler: ‘Apple clang version 14.0.0 (clang-1400.0.29.202)’
using SDK: ‘’
clang -arch x86_64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Libra
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/S
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
In file included from /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/R
/Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library/RcppEigen/include/Eigen
#include <cmath>
         ^~~~~~~
1 error generated.
make: *** [foo.o] Error 1
```

Test Data

If the statistical model cannot recreate the parameters (within sampling variance) behind simulated data that exactly matches the model, then something is wrong. In reality, timing and spread will be responses to river flow events, clearly so for WCVI, less so for Fraser River with protracted freshwater migration. Big rivers still have habitat events that control salmon migrations: high water temperatures block entry to Okanagan River from Columbia River for Osoyoos Lake sockeye.

Least realistic: proportion of run for each river is constant across all years. That involves Direchelet priors for proportions (a simplex, sum is 1). Assumes essentially one population spread across rivers instead of a population in each river. That implies rivers have independent population dynamics, run proportions will change. As tendencies to synchronize population dynamics, rivers share both freshwater and marine survival patterns and events, and spawners stray among rivers. This model is merely a starting point.

```
#|label: test_data_multi_Gauss
# data and initial guesses for testing stan_MultiGauss
#  dat[] will have 10 items:
# int<lower=0>        np;      // length of vector of priors,
#   vector [np]      prior;      // does not include river_prop_prior
#  int<lower=0>   n_river;      // how many rivers
#  int<lower=0>    n_year;      // how many years
#   int<lower=0>     n_data;      // how many data
#  vector [n_data]      x;      // day of year for each observation
#  vector [n_data]      y;      // estimate of abundance for each day in x
#  int  river_id [n_data];      // id of river in data. integer array
#  int  year_id  [n_data];      // id of year
#  vector [n_river] river_prop_prior; // Dirichelet, even when vector of ones
```

```r
# setup to simulate salmon abundance (before observing)
n_river       <-   5
n_year        <-  20
n_day         <-   6  # surveys in one river in one year.
timing        <- 280 # day of year , background mean timing
spread        <-   7 # days, background spread of timing
year_timing_sd  <- 4
year_spread_sd  <- 1
river_timing_sd <- 4
river_spread_sd <- 1
year_timing  <- rnorm(n_year,  0, year_timing_sd)  #negative OK here
year_spread  <- rnorm(n_year,  0, year_spread_sd)
river_timing <- rnorm(n_river, 0, river_timing_sd)
river_spread <- rnorm(n_river, 0, river_spread_sd)
# run size as log-normal (positive only)
   #exp8:12 ~ 3,000  8,000 22,000 60,000 163,000
   # sigma = 1 typically results in 1k to 250k
   # Stan: run = lognormal(log(prior[5]), prior[6] );
run  <- rnorm(n_year, 10, 1) %>% exp
# proportion of run in each river (same all years)
river_prop <- runif(n_river,1,10)
river_prop <- river_prop/sum(river_prop)

cat('background timing =', timing, ' background spread =', spread,
    '\n year_timing = \n ',  round(year_timing, 2),
    '\n year_spread = \n ',  round(year_spread, 2),
    '\n river_timing = \n ', round(river_timing,2),
    '\n river_spread = \n ', round(river_spread,2),
    '\n run = \n' , run,
    '\n river_prop = \n' , round(river_prop,3), '\n');
```

```
background timing = 280  background spread = 7
 year_timing =
  7.26 -3.76 3.99 -3.58 4.44 0.35 0.82 -0.86 -5.15 2.33 -5.86 2.12 -8.26 4.91 -5.07 -6.48 -5
 year_spread =
  -0.85 0.27 -0.47 0.54 0.36 -0.61 1.97 -0.83 -0.95 1.05 1.77 -0.64 0.02 2.4 1.23 0.5 0.46 -0
 river_timing =
  0.05 5.44 -0.35 1.18 -8.09
 river_spread =
  1.38 -0.81 -0.09 -0.53 -0.12
 run =
```

```
 29165.42 112421.4 31397.13 63293.02 19991.98 9672.998 2862.282 48414.4 8726.231 17251.23 95
 river_prop =
 0.125 0.48 0.268 0.054 0.074
```

```
# set day of observation in each river, always the same.
from <- timing-2*spread;
to <- timing + 2*spread;
day  <-  floor(seq(from,to, length.out=n_day)); # day of year for obs.
day_obs  <- rep(day, n_year*n_river); # in every river every year.
# create indices into x and y for river and year
# order of data: year has river has day.
n_data <- n_day*n_river*n_year
year_index  <- rep(1:n_year, each= n_river*n_day);
# 60 of 1, 60 of 2,. length is 600
river_index <- rep(rep(1:n_river, each=n_day), n_year);
# y is simulated observations. order is year, river, day.
# a 3D array would be better.
y=numeric(n_year*n_river*n_day);
j_obs <- 0 # index into data
for (j_year in 1:n_year){
    t_y <- timing + year_timing[j_year]
    s_y <- spread + year_spread[j_year]
    r_y <- run[j_year]
     for (j_river in 1:n_river){
         t <- t_y + river_timing[j_river];
         s <- s_y + river_spread[j_river];
         r <- r_y * river_prop[j_river];
         for (j_day in 1:n_day ){ # day same, all years and rivers
             j_obs <- j_obs + 1;
             y[j_obs] <- r * dnorm(day[j_day], t, s); # true abundance
         }
     }
}

sigma_sampling = 0.2
y_obs <- y * rnorm(y, 1, sigma_sampling) # sampling error is proportional.
y_obs[ y_obs < 1] = 0 # if < 1 observed, then zero
a <- data.frame (year_index, river_index, Day = day_obs, Truth = round(y,2), Observed = roun
print(a[c(1,n_data), ])
```

```
    year_index river_index Day Truth Observed
1            1           1 266  3.49     3.36
```

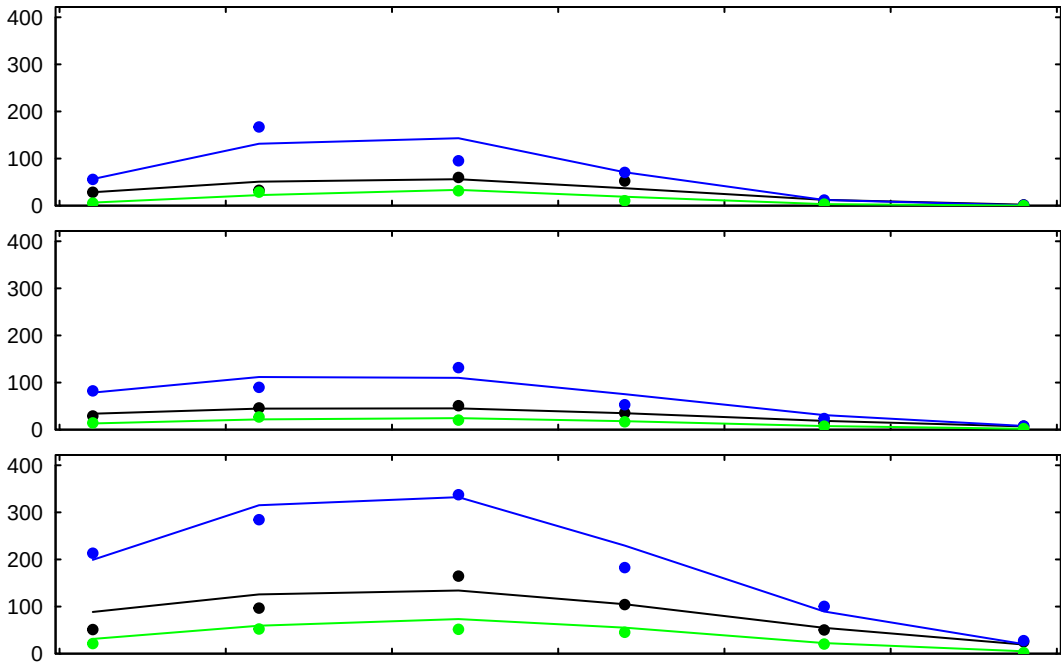
```
600          20            5 294 11.63     8.58
```

plot a sample of years and rivers

```
#|label: plt_sample_years
#years <-c(1,2,3)
#rivers <- c(1,2,3)
years  <- sample(1:n_year, 3) %>% sort
rivers <- sample(1:n_river, 3) %>% sort
cat('years:', years, '  rivers:', rivers, '\n')
```

```
years: 9 11 15   rivers: 1 3 4
```

```
j = which((a[,'year_index'] %in% years)&(a[,'river_index'] %in% rivers))
b=a[j,] # subset
ymax <- 1.25* max(b[,4:5]) # largest of true or observed

par(las=1,tcl=0.2, pch=19, yaxs="i", mgp=c(0,.5,0), mar=c(1,2,0,0), mfrow=c(length(years),1)
colr=c("black","blue","green","orange","red")

for (j in 1:length(years)){      # how many selected years
    plot( x=day, y=day, type='n', ylim=c(0,ymax), xaxt="n",ylab='',xlab='');   axis(1,labels=
    for(k in 1:length(rivers)){  # how many selected rivers
        m0 = (j-1)*length(rivers)*n_day +
             (k-1)*n_day + 1
        m  = m0:(m0+n_day-1)
        y=b[m,'Truth']
        lines(b[m,'Truth']    ~day, col=colr[k])
        points(b[m,'Observed']~day, col=colr[k])
    }
}
```




```
#   k <- 0;
#   for(j in seq(0,n_river*n_day-1,n_day)){  # 1  7 13 19 25 ...
#     k=k+1;
#     lines( b[j:(j+n_day), 3], b[j:(j+n_day), 4], col=colr[k] ); # 6 observations
#     points(b[j:(j+n_day), 3], b[j:(j+n_day), 5], col=colr[k] );
#   }
# }
#
```

Define prior knowledge

```
#|label: MG_priors
# ----- priors from Stan, defined in R --------
prior <- numeric(12);

# (1) background, mean for all years and rivers.
#   timing ~ normal(prior[1], prior[2]);   // background
prior[1:2] <- c(timing, 7);
#   spread ~ normal(prior[3], prior[4]);   // background
prior[3:4] <-c(spread, 3); # too tight, loosen later
```

```r
# (2) spawner total abundance by year.
#   run ~ lognormal(prior[5], prior[6]);
prior[5:6] <- c(10., 1.)

# (3) year and river effects
# year_timing   ~ normal(0,prior[7]);  // shared all years
prior[7] <- year_timing_sd
# year_spread   ~ normal(0,prior[8]);
prior[8] <- year_spread_sd
# river_timing  ~ normal(0,prior[9]); // shared all rivers
prior[9] <- river_timing_sd
# river_spread  ~ normal(0,prior[10]);
prior[10] <- river_spread_sd

# (4) sampling error
#  sigma         ~ gamma(prior[11], prior[12]);  // wide
# shape and scale, 0 at 0, max at 0.1, 1-pgamma(1,1.5,8)=0.001134
prior[11:12] <- c(1.5, 8.0)

# (5) distribution of total run across rivers.
#  river_prop    ~ dirichlet(river_prop_prior); // separate data vector
river_prop_prior <- rep(1.0, n_river);

names(prior) <- c('timing_m', 'timing_sd', 'spread_m', 'spread_sd',
    'run_m', 'run_sd',
    'year_timing', 'year_spread_', 'river_timing', 'river_spread',
    'sigma_gamma_shape', 'sigma_gamma_scale');
print("priors"); prior %>% round(1) %>% print;
```

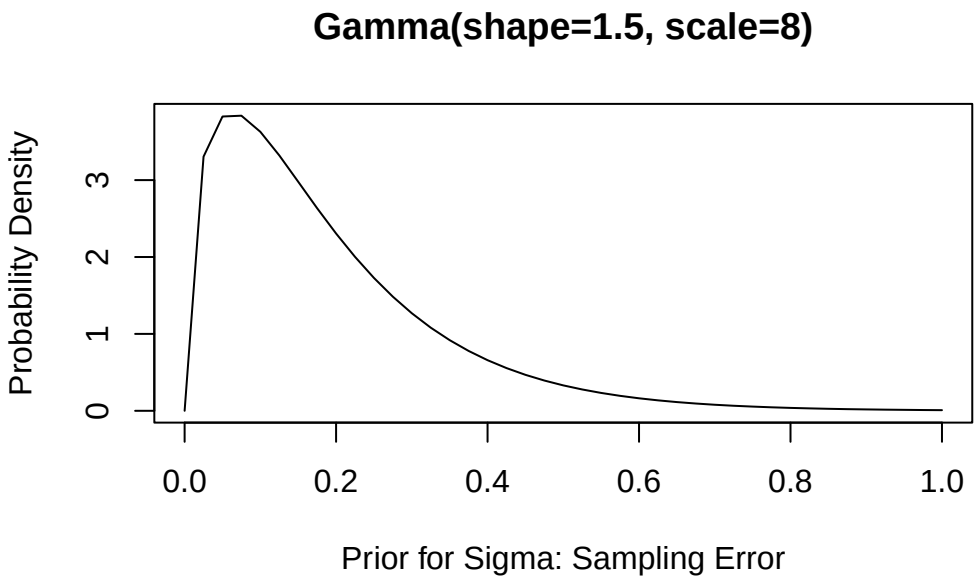
```
[1] "priors"

         timing_m         timing_sd          spread_m         spread_sd
            280.0               7.0               7.0               3.0
            run_m            run_sd       year_timing      year_spread_
             10.0               1.0               4.0               1.0
     river_timing      river_spread sigma_gamma_shape sigma_gamma_scale
              4.0               1.0               1.5               8.0
```

```r
print("river_prop_prior"); print(river_prop_prior);
```

```
[1] "river_prop_prior"
```

```
[1] 1 1 1 1 1
```

```
# sigma is proportional sampling error, prior is gamma(shape, scale)
x=seq(0,1,0.025);y= dgamma(x, 1.5, 8);
plot(x,y, type='l', xlab='Prior for Sigma: Sampling Error',ylab='Probability Density', main=
```




```
dat <- list(n_prior=12, prior=prior, n_year=n_year, n_river=n_river, n_data=n_data, day_obs=
#print("This is dat:");
# print(dat);
```

Sampling stan_MultiGauss

```
#|label: sampling_MultiGauss
#|cache: true
# library(rstan)
s1 <- sampling( stan_MultiGauss, data=dat, chains=4, iter=4000, cores=4, verbose = F)
```

print sampling

```
#|label: prt_sampling_s1
print (s1);
```

```
Inference for Stan model: file1261b7bdaf481.
4 chains, each with iter=4000; warmup=2000; thin=1;
post-warmup draws per chain=2000, total post-warmup draws=8000.

                     mean se_mean      sd      2.5%       25%       50%
timing             279.07    0.04    1.94    275.30    277.76    279.10
spread               7.10    0.01    0.55      6.00      6.73      7.10
run[1]           27226.20   19.72 1782.36  23901.22  26025.23  27159.25
run[2]          109543.98   19.99 1776.37 106050.52 108336.97 109539.35
run[3]           33237.06   15.62 1486.62  30430.21  32223.54  33203.07
run[4]           57758.42   17.09 1628.04  54570.52  56667.49  57759.13
run[5]           18119.61   18.93 1590.73  15143.81  17035.86  18049.24
run[6]           11479.47   13.35 1328.65   8959.59  10575.87  11443.00
run[7]            3100.83   11.04 1040.16   1236.92   2364.16   3037.60
run[8]           48009.22   14.87 1425.82  45301.64  47040.69  48001.13
run[9]            9927.89   12.93 1233.70   7614.62   9074.57   9899.90
run[10]          16100.53   15.16 1441.45  13368.97  15122.19  16071.28
run[11]           9273.69   13.72 1331.42   6685.91   8349.49   9253.30
run[12]           9262.33   13.09 1330.35   6761.98   8335.36   9225.77
run[13]          17063.97   14.16 1384.04  14430.97  16136.52  17021.85
run[14]          32701.68   19.59 1921.99  29043.37  31397.35  32643.00
run[15]          26336.18   16.72 1537.61  23362.86  25296.76  26319.19
run[16]          43548.85   15.34 1494.18  40626.29  42551.95  43547.35
run[17]          17586.39   15.00 1389.88  14912.74  16642.36  17565.74
run[18]          21356.11   13.89 1359.32  18739.25  20444.06  21338.55
run[19]          63049.87   14.64 1374.55  60299.77  62142.96  63055.36
run[20]          59609.63   18.99 1889.69  55963.12  58316.50  59574.34
year_timing[1]       7.24    0.03    0.98      5.36      6.59      7.23
year_timing[2]      -4.18    0.03    0.91     -5.91     -4.80     -4.19
year_timing[3]       5.38    0.03    0.94      3.57      4.74      5.37
year_timing[4]      -1.84    0.03    0.92     -3.62     -2.47     -1.84
year_timing[5]       4.52    0.03    1.12      2.38      3.76      4.51
year_timing[6]       1.01    0.03    1.24     -1.38      0.18      1.01
year_timing[7]       0.24    0.03    3.15     -5.79     -1.81      0.17
year_timing[8]       1.05    0.03    0.92     -0.73      0.44      1.04
year_timing[9]      -3.65    0.03    1.20     -6.01     -4.47     -3.64
year_timing[10]      3.33    0.03    1.10      1.20      2.59      3.32
year_timing[11]     -4.42    0.03    1.50     -7.33     -5.42     -4.43
year_timing[12]      2.76    0.03    1.39      0.01      1.82      2.76
year_timing[13]     -7.25    0.03    1.06     -9.29     -7.95     -7.26
year_timing[14]      4.75    0.03    1.00      2.84      4.07      4.75
year_timing[15]     -4.75    0.03    1.00     -6.67     -5.42     -4.76
year_timing[16]     -5.80    0.03    0.93     -7.60     -6.44     -5.81
```

year_timing[17]	-4.20	0.03	1.10	-6.31	-4.93	-4.18
year_timing[18]	0.18	0.03	1.00	-1.79	-0.49	0.17
year_timing[19]	1.07	0.03	0.91	-0.69	0.45	1.06
year_timing[20]	4.22	0.03	0.95	2.40	3.58	4.21
year_spread[1]	-0.96	0.01	0.50	-1.91	-1.30	-0.98
year_spread[2]	0.56	0.01	0.28	0.00	0.37	0.56
year_spread[3]	-0.96	0.01	0.39	-1.70	-1.22	-0.97
year_spread[4]	0.00	0.01	0.34	-0.67	-0.24	-0.01
year_spread[5]	0.34	0.01	0.65	-0.87	-0.10	0.32
year_spread[6]	0.01	0.01	0.72	-1.33	-0.48	-0.01
year_spread[7]	0.29	0.01	0.96	-1.55	-0.36	0.28
year_spread[8]	-0.85	0.01	0.32	-1.48	-1.06	-0.85
year_spread[9]	-0.65	0.01	0.74	-2.03	-1.18	-0.68
year_spread[10]	-0.14	0.01	0.63	-1.31	-0.58	-0.17
year_spread[11]	0.44	0.01	0.79	-1.09	-0.09	0.41
year_spread[12]	0.03	0.01	0.78	-1.39	-0.52	0.00
year_spread[13]	-0.13	0.01	0.57	-1.19	-0.51	-0.14
year_spread[14]	1.13	0.01	0.54	0.11	0.76	1.12
year_spread[15]	0.67	0.01	0.50	-0.28	0.33	0.66
year_spread[16]	0.18	0.01	0.35	-0.50	-0.05	0.18
year_spread[17]	0.34	0.01	0.55	-0.72	-0.05	0.33
year_spread[18]	-0.16	0.01	0.49	-1.11	-0.50	-0.18
year_spread[19]	-1.20	0.01	0.29	-1.76	-1.40	-1.20
year_spread[20]	1.12	0.01	0.36	0.42	0.87	1.11
river_timing[1]	0.53	0.03	1.76	-2.94	-0.65	0.53
river_timing[2]	5.51	0.03	1.73	2.10	4.34	5.51
river_timing[3]	0.05	0.03	1.74	-3.36	-1.10	0.05
river_timing[4]	1.38	0.03	1.78	-2.10	0.19	1.38
river_timing[5]	-7.86	0.03	1.80	-11.44	-9.07	-7.86
river_spread[1]	1.32	0.01	0.58	0.19	0.93	1.31
river_spread[2]	-1.02	0.01	0.49	-1.98	-1.35	-1.03
river_spread[3]	0.05	0.01	0.50	-0.92	-0.30	0.04
river_spread[4]	-0.63	0.01	0.62	-1.79	-1.06	-0.65
river_spread[5]	0.27	0.01	0.67	-0.99	-0.18	0.26
river_prop[1]	0.12	0.00	0.00	0.11	0.12	0.12
river_prop[2]	0.47	0.00	0.01	0.46	0.47	0.47
river_prop[3]	0.27	0.00	0.00	0.26	0.27	0.27
river_prop[4]	0.06	0.00	0.00	0.05	0.06	0.06
river_prop[5]	0.07	0.00	0.01	0.06	0.07	0.07
sigma	57.97	0.01	1.19	55.73	57.16	57.96
lp__	-3477.63	0.12	6.56	-3491.32	-3481.81	-3477.29

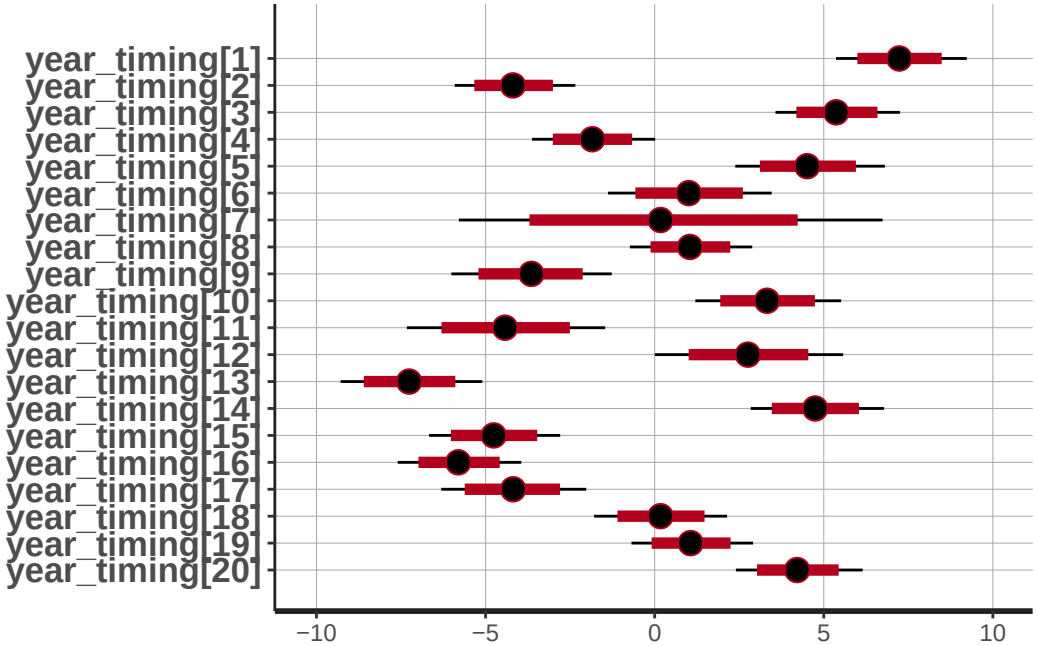
	75%	97.5%	n_eff	Rhat
timing	280.38	282.88	2196	1

spread	7.47	8.14	2916	1
run[1]	28343.81	30916.96	8165	1
run[2]	110739.35	113076.30	7900	1
run[3]	34208.71	36194.35	9059	1
run[4]	58830.95	60968.62	9076	1
run[5]	19176.22	21324.80	7060	1
run[6]	12365.45	14148.41	9902	1
run[7]	3773.99	5306.62	8883	1
run[8]	48977.37	50825.55	9200	1
run[9]	10768.83	12374.07	9110	1
run[10]	17027.74	19063.23	9043	1
run[11]	10148.07	11941.97	9421	1
run[12]	10135.33	11932.80	10330	1
run[13]	17984.98	19885.36	9549	1
run[14]	33933.14	36675.97	9623	1
run[15]	27370.84	29397.63	8453	1
run[16]	44536.08	46536.20	9488	1
run[17]	18508.86	20404.46	8586	1
run[18]	22265.48	24047.72	9583	1
run[19]	63947.58	65752.92	8817	1
run[20]	60856.98	63448.10	9905	1
year_timing[1]	7.87	9.23	1256	1
year_timing[2]	-3.59	-2.34	1111	1
year_timing[3]	6.00	7.26	1174	1
year_timing[4]	-1.22	0.02	1122	1
year_timing[5]	5.24	6.81	1562	1
year_timing[6]	1.83	3.46	2048	1
year_timing[7]	2.27	6.74	10437	1
year_timing[8]	1.66	2.89	1130	1
year_timing[9]	-2.86	-1.27	1867	1
year_timing[10]	4.06	5.51	1558	1
year_timing[11]	-3.42	-1.46	2939	1
year_timing[12]	3.68	5.57	2435	1
year_timing[13]	-6.55	-5.10	1473	1
year_timing[14]	5.40	6.78	1310	1
year_timing[15]	-4.08	-2.79	1283	1
year_timing[16]	-5.19	-3.94	1164	1
year_timing[17]	-3.47	-2.02	1500	1
year_timing[18]	0.86	2.14	1304	1
year_timing[19]	1.68	2.91	1121	1
year_timing[20]	4.85	6.15	1190	1
year_spread[1]	-0.64	0.06	5166	1
year_spread[2]	0.74	1.11	2823	1

```
year_spread[3]      -0.71     -0.19  4021    1
year_spread[4]       0.24      0.68  3443    1
year_spread[5]       0.76      1.69  6258    1
year_spread[6]       0.48      1.48  8674    1
year_spread[7]       0.94      2.18 10744    1
year_spread[8]      -0.63     -0.22  3250    1
year_spread[9]      -0.16      0.85  7860    1
year_spread[10]      0.28      1.18  7413    1
year_spread[11]      0.96      2.07  9565    1
year_spread[12]      0.54      1.62  9111    1
year_spread[13]      0.24      1.04  6727    1
year_spread[14]      1.49      2.23  6261    1
year_spread[15]      1.00      1.65  5851    1
year_spread[16]      0.41      0.87  3586    1
year_spread[17]      0.71      1.45  7616    1
year_spread[18]      0.15      0.83  5372    1
year_spread[19]     -1.01     -0.63  2837    1
year_spread[20]      1.36      1.83  4050    1
river_timing[1]      1.71      4.00  3040    1
river_timing[2]      6.69      8.90  2911    1
river_timing[3]      1.23      3.44  2903    1
river_timing[4]      2.57      4.87  2944    1
river_timing[5]     -6.63     -4.38  3071    1
river_spread[1]      1.71      2.46  5038    1
river_spread[2]     -0.70     -0.05  3822    1
river_spread[3]      0.38      1.03  3901    1
river_spread[4]     -0.22      0.63  5810    1
river_spread[5]      0.72      1.64  5762    1
river_prop[1]        0.13      0.13  8852    1
river_prop[2]        0.48      0.49  6280    1
river_prop[3]        0.27      0.28 11133    1
river_prop[4]        0.06      0.07  7976    1
river_prop[5]        0.08      0.08  6675    1
sigma               58.76     60.31 10236    1
lp__             -3472.98  -3465.83  3135    1

Samples were drawn using NUTS(diag_e) at Tue Aug  6 14:01:57 2024.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).
```

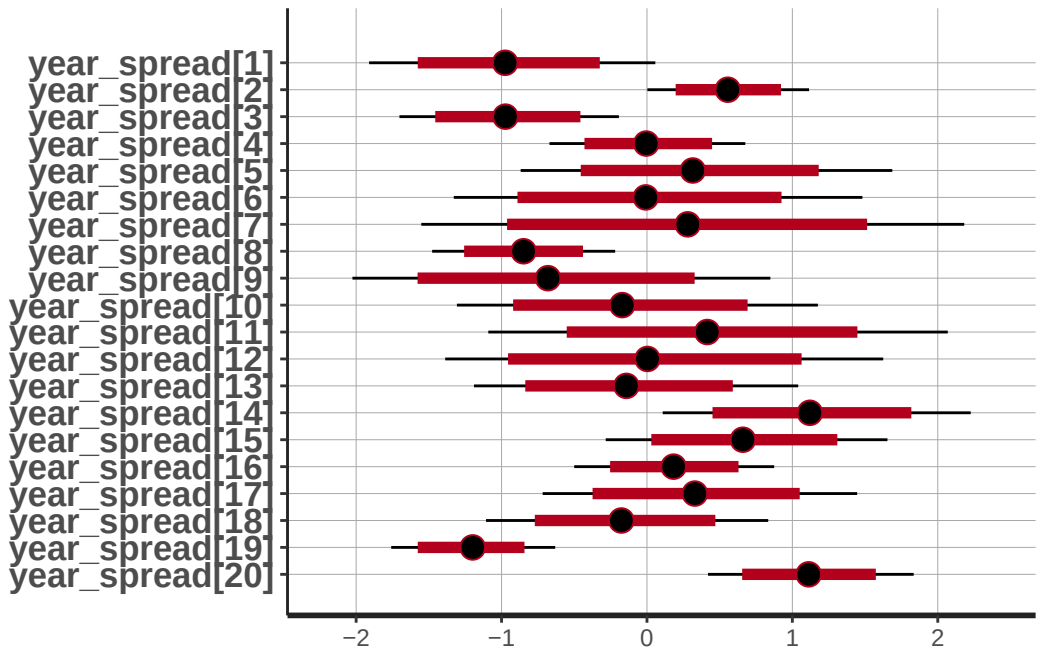
```r
plot(s1, pars=c('year_timing'));
```

```
ci_level: 0.8 (80% intervals)

outer_level: 0.95 (95% intervals)
```



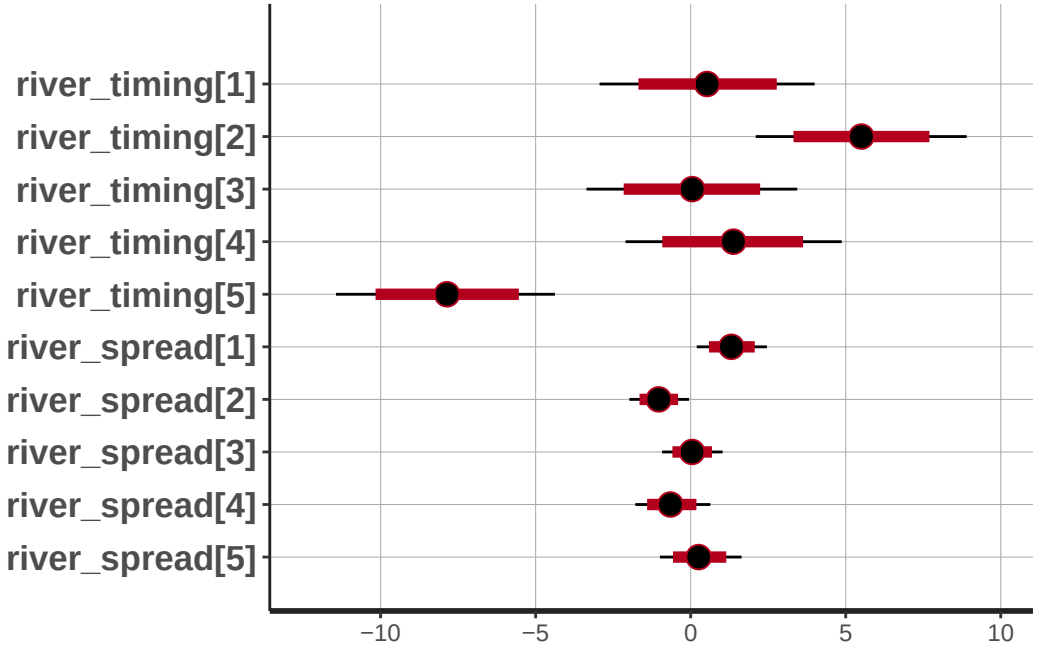

```r
plot(s1, pars=c('year_spread'));
```

```
ci_level: 0.8 (80% intervals)
outer_level: 0.95 (95% intervals)
```




```
plot(s1, pars=c('river_timing','river_spread'))
```

```
ci_level: 0.8 (80% intervals)
outer_level: 0.95 (95% intervals)
```



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